

Rajiv V. Joshi

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Education

Carnegie Mellon University – Pittsburgh, PA

May 2026

Master of Science in Mechanical Engineering - Research | GPA: 4.0/4.0

- Relevant Coursework: ADV Mechatronics (TA), Fundamentals of Mechatronic Design (TA), Modern Control Theory, Intro to Robot Learning, ML and AI for Engineers, Computer Vision for Engineers, Engineering Haptic Interfaces

Virginia Tech – Blacksburg, VA

May 2024

B.S. Mechanical Engineering – Major: Robotics and Mechatronics | GPA: 3.64/4.0

B.S. Mechanical Engineering – Major: Mechanical Engineering

Skills

Hardware: Cross System Integration, Electro-Mechanical Debugging, PCB/Circuit Design, CAD (SolidWorks), FDM Printing

Programming: Python, C/C++, MATLAB, Linux, Data Processing + Analysis, Git/GitHub

Embedded Systems: BLDC Motors, Serial Protocols (SPI, I2C, CAN), Jetson, Raspberry Pi, Arduino, STM32, KiCAD

Research Experience

Knee Exoskeleton: Master's Research Project, Carnegie Mellon University – Pittsburgh, PA

04/2025 – Present

- Coordinated and led a team of 5 students in research, design, and manufacturing an open-source knee exoskeleton capable of delivering 18 Nm of torque to knee joints for people suffering from osteoarthritis or gait impairment
- Established and wrote a modular real-time firmware architecture abstracting low-level motor + multi-sensor commands from high-level data-driven controllers enabling fast and efficient performance with system modularity
- Formulated and carried out experimental protocol for validating device performance across multiple subjects and performed data post processing, in python, to analyze experimental outcomes
- Implemented a data-driven imitation learning controller that successfully reduced leg muscle activation by 11% and energy expenditure by 12% in able bodied individuals

Hip Exoskeleton: Master's Research, Carnegie Mellon University – Pittsburgh, PA

05/2024 – 01/2026

- Completed electro-mechanical system integration of soldered PCB's, 3D printed hardware, and sensor + actuator firmware protocols; tested performance through numerous experiments wearing the device with various controllers
- Debugged multiple hardware and firmware faults using an oscilloscope, DMM and Logic Analyzer to visualize motor CAN, sensor signals, and circuit continuity to rapidly diagnose problems and iterate/manufacture immediate solutions
- Reverse engineered multiple COTS exoskeletons analyzing how hardware interface variation affects sensor stability
- Redesigned PCB schematics, in KiCAD, to integrate CAN bus and I2C protocols, in Python and C/C++, with embedded system peripherals, reducing connections count from 8 to 4 while maintaining signal integrity and minimized latency
- Optimized multiple hip exoskeletons hardware, through referencing human biomechanics, in SolidWorks for improved fit and torque transfer yielding 25% reduced system weight with 5% reduction in energy expenditure

Projects

Dexkit: Engineering Haptic Interfaces (16880), Carnegie Mellon University – Pittsburgh, PA

01/2025 – 05/2025

- Developed an open-source haptic fingers device in a team of 6, for teaching robot kinematics and haptic feedback
- Implemented event-driven code on an STM32 MCU controlling 4 low torque motors by calculating forward and inverse kinematics of a capstan mechanism

Whac-a-Wacko: ADV Mechatronics (16878), Carnegie Mellon University – Pittsburgh, PA

08/2024 – 12/2024

- Collaborated to create a bare metal mechatronic game system having 3 user inputs + 3 feedback + 2 game modes
- Developed event-driven code handling GPIO functions, timers, event queues, PWM signals, and motor control to achieve desired game performance using C on an STM32

Solar Car Dynamics Lead: Undergraduate Capstone Project, Virginia Tech – Blacksburg, VA

09/2021 – 08/2024

- Placed 12th/24th as a first-time team in the 2024 American Solar Car Challenge (ASC) Formula Sun Grand Prix
- Led Design and assembly of a double wishbone suspension and off-set Ackerman steering geometry; performed 5g impact analysis using ANSYS Mechanical to meet ASC regulations
- Managed cross team system integration using RACI tables and Action Item spreadsheets to identify critical bottlenecks and suggest alternative solutions to keep project on track for competition deadlines
- Created standardized documentation for team organization and knowledge transfer, co-taught weekly workshops to prepare team for long-term success; delivered monthly PDRs/CDRs to ASC and design reviews to capstone professors